Executive Business Report

Sales Forecasting & Demand Intelligence System

Executive Summary :

This project was developed to help improve inventory planning and sales forecasting for a retail business. By analyzing four years of historical sales data, the system identifies sales trends, predicts future demand, detects unusual sales events, and groups products based on their demand patterns. The results help the business maintain the right inventory levels, reduce unnecessary storage costs, and minimize lost sales caused by stock shortages. Overall, the system provides a practical decision-support tool for supply chain planning and inventory management.

Key Findings :

Analysis of historical sales data revealed several important business insights:

- **Technology** was the highest revenue-generating product category, contributing the largest share of total sales.
- The **West** region showed the strongest and most consistent sales growth over the four-year period.
- Average shipping time was approximately **3–4 days**, indicating efficient order fulfillment across most regions.
- Monthly sales consistently increased during the final quarter of each year, showing a clear seasonal pattern likely driven by festive shopping and year-end promotions.
- Sales generally increased over time, indicating overall business growth.

Three-Month Sales Forecast :

The forecasting system predicts that sales will remain strong over the next three months with stable growth.

Month	Forecasted Sales
Month 1	86,465.82
Month 2	86,506.77
Month 3	84,327.28

The forecast suggests that demand will remain relatively stable over the next quarter, with only minor month-to-month variation. Based on historical trends, the business should maintain sufficient inventory during this period while closely monitoring high-demand categories to prevent stock shortages.

Major Unusual Sales Events :

The system identified several weeks where sales were significantly different from normal business activity. The three most important anomalies include:

1. March 22, 2015

Sales increased sharply, likely due to promotional campaigns, bulk customer purchases, or seasonal buying behavior.

2. November 18, 2018

A significant sales spike was observed during the festive shopping period, likely influenced by discount campaigns and increased customer demand.

3. December 2, 2018

One of the highest sales weeks in the dataset, most likely driven by year-end holiday shopping and promotional offers.

Identifying these unusual events helps planners prepare inventory and logistics for future demand surges.

Product Demand Segmentation :

Products were grouped into four demand segments to support better inventory planning.

Demand Segment	Recommended Inventory Strategy
High Volume, Stable Demand	Maintain consistently high inventory levels to avoid stock shortages.
Low Volume, Stable Demand	Keep moderate inventory levels and replenish stock periodically to reduce storage costs.
High Value, High Volatility	Monitor demand closely and maintain safety stock because demand fluctuates significantly.
Growing Demand	Gradually increase inventory to support rising customer demand and future sales growth.

This segmentation enables more efficient stock allocation instead of using the same inventory policy for all products.

Business Recommendations :

1. Increase Inventory for High-Demand Products

High-volume product groups consistently generate strong sales. Maintaining higher inventory for these products will reduce lost sales caused by stock shortages.

2. Prepare Inventory Before Peak Seasonal Months

Historical data shows recurring sales increases during the final months of each year. Increasing inventory before these periods will improve customer satisfaction and revenue opportunities.

3. Monitor Sales Anomalies Continuously

Unusual sales spikes often indicate successful promotional campaigns or changing customer behavior. Regular monitoring allows the business to respond quickly by adjusting inventory, staffing, and supply chain operations.

Risk / Limitation :

The forecasting system relies primarily on historical sales data. Unexpected external events such as economic changes, supplier disruptions, new competitors, or major promotional campaigns that are not reflected in past data may reduce forecasting accuracy. Regular model updates with the latest sales information are recommended to maintain reliable predictions.

Conclusion :

The developed sales forecasting system provides valuable support for inventory planning, demand forecasting, and business decision-making. By combining historical sales analysis, future demand prediction, anomaly detection, and product segmentation, the system helps reduce inventory costs, improve stock availability, and support data-driven supply chain decisions. With regular updates and continuous monitoring, the solution can serve as a practical tool for improving operational efficiency and long-term business performance.

By :
Name : Sarthak Singh
Email : sarthakksingh220@gmail.com